Weekly Project Report

Web-Based Certificate Scanning and Error Detection System

1. Tasks Completed, In Progress, and Planned

Status	Task	Progress
Completed	Project requirements and problem understanding	Completed
Completed	Initial analysis of scanned certificate verification	Completed
In Progress	Evaluation of visual/pixel-based certificate comparison	In progress
In Progress	Research on image alignment and preprocessing	In progress
In Progress	Evaluation of OCR for Hindi and English text	In progress
Planned	Prototype for scanned-certificate comparison	Planned
Planned	Difference-map and heatmap generation	Planned
Planned	Human-assisted verification workflow	Planned

2. Problems

• Hindi and English text are present in the certificates, making text-based OCR verification more challenging.

• Scanned documents may contain rotation, translation, scaling, uneven lighting, blur, compression artifacts, and scanner noise.

• Direct pixel-by-pixel comparison can produce false differences unless the scanned document is first aligned and normalized.

• The certificate verification process has a zero-tolerance requirement, so automated detection should assist a human reviewer rather than be treated as the final authority.

• Further investigation is required to distinguish genuine printing/layout errors from differences caused by scanning conditions.

3. Time Spent by Team Members

Team Member	Task	Hours
Member 1	Project requirement analysis	1.5
Member 2	OCR / Hindi text research	2
Member 3	Image-processing research	1
Member 4	Frontend / interface work	1
Member 5	Backend / system research	1

4. Last Meeting with the PoC

Date: 31 – Aug – 2026

Discussion: About Hindi and English text in certificates, Zero error tolerance, verification of documents, Heavily time consuming manual process and Human assisted verification workflow.

5. Date of Current Build

Build date: No current build

6. Results of Tests on the Build

Metric	Result
Total number of tests	0
Successful tests	0
Unsuccessful tests	0

7. Defects Found in QA Process

No formal QA testing has been conducted yet.